Research Statement

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1. Summary of my results

I aim to study symplectic varieties, which are important building blocks for varieties whose canonical divisor is trivial. We will assume that the symplectic varieties admit Lagrangian fibrations; this is a mild condition granting the symplectic varieties extra structure.

Definition 1.1. A normal variety X is **symplectic** if it admits a non-degenerate symplectic form $\sigma \in H^0(X^{\mathrm{sm}}, \Omega^2_{X^{\mathrm{sm}}})$ which extends holomorphically onto a resolution of singularities $\pi : Y \to X$. In this setting, a **Lagrangian fibration** is a proper surjective morphism $f : X \to B$ with connected half-dimensional fibers, such that $\sigma|_{X_b} = 0$ for any fiber X_b.

We will need to assume that X is compact for most of our results. Additionally, we will want X to be "irreducible" in a sense, which will be made more precise in §2.

Definition 1.2. A compact symplectic variety X is **primitive symplectic** if it satisfies $H^1(X, \mathcal{O}_X) = 0$ and σ is unique up to scaling, i.e. $H^0(X, \Omega^2_X) = \mathbb{C}\sigma$.

By Schwald [Sch20], when X is a primitive symplectic variety admitting a Lagrangian fibration $f : X \to B$, the smooth fibers X_b are Abelian varieties. My main results further classify the structure of the fibers.

Theorem 1.3 ([Var26b])**.** *Let X be a primitive symplectic variety of dimension $2n$ admitting a Lagrangian fibration $f : X \to B$. If f is not isotrivial, the Mumford–Tate group of the very general fiber X_b is isogenous to $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^{n/h}$.*

Recall that an Abelian variety is Hodge-generic if it admits no extra Hodge tensors. In this case, it is simple and carries no extra symmetries.

Corollary 1.4 ([Var26b])**.** *The very general fiber X_b is isogenous to a product of Abelian varieties $A_1 \times \cdots \times A_{n/h}$, where each A_i is Hodge-generic and A_i is not isogenous to A_j for $i \neq j$.*

The most well-studied class of primitive symplectic varieties consists of hyper-Kähler manifolds, which should be thought of as the natural generalization of K3 surfaces.

Definition 1.5. A compact Kähler manifold X is **hyper-Kähler** if X is simply connected and satisfies $H^0(X, \Omega^{2p}_X) = \mathbb{C}\sigma^p, \quad H^0(X, \Omega^{2p+1}_X) = 0$.

Theorem 1.6 ([Var26b])**.** *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds such that all members $X \in \mathcal{X}$ have second Betti number $b_2(X) \geq 7$. Choose X to be general among the hyper-Kähler manifolds admitting a Lagrangian fibration, $f : X \to B$. If X_b is a very general fiber of f, then its Mumford–Tate group $\mathbf{MT}(X_b)$ is isogenous to GSp_{2n}, i.e. X_b is Hodge-generic.*

These results largely came about through understanding the monodromy of the family f. To be more precise, let $f : X^\circ \to B^\circ$ be the restriction of f to its regular points, and let $V = R^1 f_* \mathbb{Z}_{X^\circ}$, viewed as a local system.

Theorem 1.7 ([Var26a])**.** *Suppose that X is primitive symplectic, $f : X \to B$ a Lagrangian fibration, and let $V_{\mathbb{C}} = V \otimes_{\mathbb{Z}} \mathbb{C}$ be the underlying $\mathbb{C}$-representation of $\pi_1(B^\circ)$.*

(1) *If f is not isotrivial, then $V_{\mathbb{C}}$ is irreducible.*
(2) *If f is isotrivial, then its smooth fibers are isogenous to E^n, for E an elliptic curve. Furthermore, if K is the endomorphism field of E (possibly $\mathbb{Q}$), then $V_K = U_1 \oplus U_2$ splits into two geometrically irreducible components.*

Remark 1.8. There are a few clarifications to be made regarding the results and contributions. For the non-isotrivial case (1), the later result of Theorem 1.3 determines that a finite-index subgroup of the image of $\pi_1(B^\circ)$ lies in $(\mathrm{Sp}_{2h,\mathbb{Q}})^{n/h}$, and acts on $V_{\mathbb{C}}$ via a direct sum of standard representations. Such a representation would not be irreducible, if not for the unaccounted finite part of the monodromy transitively permuting the irreducible factors. For the isotrivial case (2), the result $X_b \sim E^n$ was first proven in [KLM26], though we produced an alternative proof of this fact.

2. Background and motivation

Hyper-Kähler manifolds arise naturally in the study of K-trivial varieties. The following result is a classical theorem of Beauville–Bogomolov:

Theorem 2.1 (Beauville–Bogomolov Decomposition Theorem)**.** *Let Z be a compact Kähler manifold with $\omega_Z \simeq \mathcal{O}_Z$. Then there is a finite étale cover $\pi : Z' \to Z$ such that Z' has the form*

$$Z' = T \times \prod_i X_i \times \prod_j Y_j,$$

where T is a complex torus, the X_i are hyper-Kähler manifolds, and the Y_j are irreducible Calabi–Yau manifolds.

This theorem has since been generalized to allow for Z to have klt singularities (cf. [BGL22]); in this case, the X_i are irreducible holomorphic symplectic varieties, a condition stronger than primitive symplectic but weaker than hyper-Kähler.

Thus the problem of understanding families of K-trivial varieties boils down to understanding families of complex tori, hyper-Kählers, and irreducible Calabi–Yaus. The former is largely classical, while the latter is also somewhat well-studied, largely due to work of Friedman–Laza. In contrast, the deformation spaces of hyper-Kähler manifolds are not as well understood.

Given a polarized hyper-Kähler manifold (X, H), one can construct its deformation space as a type IV Shimura variety: take $L = H^\perp \subsetneq H^2(X, \mathbb{Z})$ with respect to the Beauville–Bogomolov–Fujiki form q, and define

$$\mathcal{D}_H = \{[\sigma] \in \mathbb{P}L_{\mathbb{C}} \mid q(\sigma, \sigma) = 0, \quad q(\sigma, \overline{\sigma}) > 0\}.$$

Then the deformation space of X is constructed as an open subset

$$\mathcal{X} \subseteq \Gamma \backslash \mathcal{D}_H = \Gamma \Big\backslash SO(2, b_2 - 3) \Big/ SO(2) \times SO(b_2 - 3) \ ,$$

where $b_2 = \dim H^2(X, \mathbb{C})$.

As such, to construct a deformation space of hyper-Kähler manifolds, it suffices to exhibit one member. There are currently four known deformation types: the infinite families $K3^{[n]}$ and Kum_n, occurring in every even dimension $2n$, and the two sporadic types $OG6$ and $OG10$, occurring in dimensions 6 and 10 respectively.

Now consider the following central result, recently proven by OpenAI [Ope26b] and by Engel–Mauri [EM26].

Theorem 2.2 (Hyper-Kähler SYZ Conjecture)**.** *Let X be a hyper-Kähler manifold and L a non-trivial nef line bundle with $q(L) = 0$. Then L is semiample, and so induces a Lagrangian fibration $f : X \to B$.*

In particular, if $b_2 \geq 5$, every deformation space $\mathcal{X}$ contains some X admitting a Lagrangian fibration. Thus one can make statements about all deformation classes of hyper-Kählers with $b_2 \geq 5$ by proving them for those admitting Lagrangian fibrations. For instance, Engel–Mauri combine Theorem 2.2 with [EFG+25] to show that there are only finitely many deformation classes of hyper-Kählers in any fixed dimension.

As such, I believe it important to understand the structure of Lagrangian fibrations. Incidentally, it turns out that they carry many constraints.

Theorem 2.3. *Let X be a hyper-Kähler manifold of dimension $2n$, $f : X \to B$ a proper surjective morphism with connected fibers, $B \neq \{p\}$. The following all hold.*

(1) [GL14] *f is a Lagrangian fibration.*
(2) [Mat99] *The smooth fibers of f are Abelian varieties.*
(3) [Mat16] *B is $\mathbb{Q}$-Fano, $\mathbb{Q}$-factorial, and klt with Picard rank 1. There is an isomorphism $\mathrm{IH}^*(B, \mathbb{Q}) \simeq H^*(\mathbb{P}^n, \mathbb{Q})$.*
(4) [Hwa08] *If B is smooth, then $B \simeq \mathbb{P}^n$.*
(5) [Bak25] *If f is not isotrivial, then it varies maximally, i.e., the period map $\mathcal{P} : B^\circ \to \mathcal{A}_n$ has an n-dimensional image.*

My results Corollary 1.4 and Theorem 1.6 strengthen (5) by understanding where $\mathcal{P}$ lands inside of $\mathcal{A}_g$. This consequently strengthens (2), by deducing the structure of the very general smooth fiber. The base B is a common area of study in this field, due to the following conjecture of Matsushita.

Conjecture 2.4 (Matsushita)**.** *Let X be a hyper-Kähler manifold, $f : X \to B$ a Lagrangian fibration. Then $B \simeq \mathbb{P}^n$.*

This conjecture was partially resolved by Hwang in (4), and was recently proven by OpenAI [Ope26a] whenever X is projective. With the base thus understood, much of the remaining information in f is carried by the period map $\mathcal{P}$, which is the focus of my results.

3. Future directions

3.1. **A geometric characterization for the Mumford–Tate splitting.** Let $f : X \to B$ be a Lagrangian fibration, and suppose the Mumford–Tate group of the very general fiber X_b is of the form

$$\mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^k, \quad k \neq 1. \tag{3.1}$$

It is not hard to see that such a Lagrangian fibration exists. For example, we can take $g : S \to \mathbb{P}^1$ to be an elliptically fibered K3 surface; the Hilbert scheme of points $X := S^{[n]}$ is hyper-Kähler, and when g is not isotrivial, X admits a Lagrangian fibration

$$f = g^{[n]} : X \to \mathbb{P}^n, \quad \mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{SL}_2)^n.$$

More generally, if we allow X to be primitive symplectic, it is easy to see that we can attain every positive integer pair (h, k) in (3.1): by Theorem 1.6, we can find a Lagrangian

fibration $g : Y \to \mathbb{P}^h$ with $\mathbf{MT}(Y_b) \sim \mathbb{G}_m \cdot \mathrm{Sp}_{2h}$, and taking the k-th symmetric power of this morphism produces

$$X = Y^{(k)} \xrightarrow{g^{(k)}} (\mathbb{P}^h)^{(k)},$$

a Lagrangian fibration realizing the Mumford–Tate group $\mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^k$.

However, notice that the base $(\mathbb{P}^h)^{(k)}$ is easily distinguished from projective space when $h \neq 1$, whereas the base of a Lagrangian fibration on a projective hyper-Kähler manifold must be $\mathbb{P}^n$ [Ope26a]. This ensures that the argument above does not provide a hyper-Kähler manifold X realizing the prescribed Lagrangian fibration $f : X \to B$.

Question 3.1. *Does there exist a hyper-Kähler manifold (or even an irreducible holomorphic symplectic variety) X admitting a Lagrangian fibration $f : X \to B$ such that $\mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^k$ for some $h \neq 1, n$?*

I expect a negative answer to this question. This would imply that if X_b is not Hodge-generic, the variation of Hodge structures underlying f can be understood in terms of variations of elliptic curves.

It is easy to exhibit such a Lagrangian fibration: as above, let $\varphi : S \to \mathbb{P}^1$ be an elliptically fibered K3 surface, and consider the map induced on the Hilbert scheme of g points: $\varphi^{[g]} : S^{[g]} \to (\mathbb{P}^1)^{[g]} \simeq \mathbb{P}^g$. The fibers of $\varphi^{[g]}$ are isogenous to $E_1 \times \cdots \times E_g$ a product of elliptic curves, as needed. We also have a cover $\pi : (\mathbb{P}^1)^g \to \mathbb{P}^g$ satisfying the following diagram:

$$\begin{array}{ccc} \underbrace{S \times \cdots \times S}_{g} & \dashrightarrow & X \\ \downarrow{\scriptstyle \varphi^g} & & \downarrow{\scriptstyle \varphi^{[g]}} \\ \underbrace{\mathbb{P}^1 \times \cdots \times \mathbb{P}^1}_{g} & \xrightarrow{\pi} & \mathbb{P}^g \end{array}$$

But one can imagine such a diagram existing more generally. Thus, one can ask if the following partial converse holds:

Question 3.2. *Let X be a hyper-Kähler manifold, $f : X \to B$ a Lagrangian fibration. Suppose $\mathbf{MT}(X_b) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^k$, for $h < n$. Does X belong to the deformation type $K3^{[n]}$?*

3.2. **Understanding the period image of related weight one variations.** My results tend to assume that X is hyper-Kähler (or primitive symplectic) because we desire some irreducibility results, the most crucial of which is Bakker's [Bak25]: f is either isotrivial or varies maximally. However, there are many ways I hope to generalize my results to less restrictive settings.

Question 3.3. *Let M be a symplectic manifold (not necessarily compact), and let $f : M \to B$ be a Lagrangian fibration. Suppose that f varies maximally; show that*

$$\mathbf{MT}(M_b) \sim \mathbb{G}_m \cdot \left(\mathrm{Sp}_{2h_1} \times \cdots \times \mathrm{Sp}_{2h_k}\right),$$

for some collection of integers h_i each dividing $\dim B$.

I believe the methods used in my prior papers should be able to accomplish this, up to potentially needing to strengthen the conditions on f a bit further. At present, I do not see how to avoid the assumption that f varies maximally in this generality. On the other hand, there are some special cases where this assumption seems unnecessary.

Recall that for C a smooth projective curve, the moduli space of polystable Higgs bundles $\mathcal{M}_{\mathrm{Dol}}(C,r)$ admits a natural Lagrangian fibration $h : \mathcal{M}_{\mathrm{Dol}}(C,r) \to \mathbb{A}$, called the Hitchin fibration. Because the fibration is constant with respect to the scaling action $(E,\theta) \mapsto (E,\lambda\theta)$, the Hitchin fibration has no hope of varying maximally. However, I can extend the results of [Var26a] to h, to show that $R^1h_*\mathbb{C}|_{\mathbb{A}^\circ}$ is an irreducible $\mathbb{C}$-local system.

Question 3.4. *Let F be a very general fiber of the Hitchin fibration h. Do we have $\mathbf{MT}(F) \sim \mathbb{G}_m \cdot (\mathrm{Sp}_{2h})^k$? Do we have $\mathbf{MT}(F) \sim \mathrm{GSp}_{2g}$?*

This may not even need a very technical argument; since F is known to be the Jacobian of a spectral curve, we may be able to apply some classical vanishing cycle results to this setting.

3.3. **Towards the hyper-Kähler Bogomolov–Mumford conjecture.** Recall the classical theorem of Bogomolov–Mumford: an ample linear series on a K3 surface contains a divisor supported on rational curves. An analogue of this theorem remains open for hyper-Kähler manifolds.

Conjecture 3.5 (Hyper-Kähler Bogomolov–Mumford Conjecture)**.** *Let X be a hyper-Kähler manifold, L an ample line bundle on X. Then there exists a uniruled divisor $D \in |L|$.*

As of now, this conjecture is only known in full for the deformation types $K3^{[n]}$ and Kum_n. Currently, I do not even have an idea on how to attack it in full generality, but Theorem 1.6 may be useful in one aspect of this conjecture: producing an abundance of rational curves.

If we start with X a general hyper-Kähler admitting a Lagrangian fibration $f : X \to B$ as in the assumptions of Theorem 1.6, the image of the period map $B^\circ \dashrightarrow \mathcal{A}_n$ will be n-dimensional. On the other hand, the product locus $\mathcal{A}_1 \times \mathcal{A}_{n-1} \subseteq \mathcal{A}_n$ has codimension $n-1$. The period image is Hodge-generic, so by Ax–Schanuel, it must meet $\mathcal{A}_1 \times \mathcal{A}_{n-1}$ in the expected dimension $n-(n-1)=1$.

In other words, there is a curve $C^\circ \subseteq B^\circ$ over which the family of Abelian varieties $X_{C^\circ} \to C^\circ$ is isogenous to a product family. It should be reasonable to show that over the boundary of C°, the elliptic part of the family degenerates into a rational curve, while the non-elliptic part remains non-degenerate. This gives rise to a uniruled algebraic cycle of dimension $n+1$ in $\mathrm{CH}^*(X)$.

Thus, we have methods of producing rational curves in the singular fibers of f. However, we want to produce rational curves on all hyper-Kähler manifolds, not simply the ones admitting Lagrangian fibrations. By Theorem 2.2, this reduces to the following question:

Question 3.6. *Let X be a hyper-Kähler manifold admitting a Lagrangian fibration f. Let $Z \subseteq X$ be a singular rational curve contained in a singular fiber of f, obtained as above. Let X' be a general deformation of X; does Z deform to a rational curve $Z' \subseteq X'$?*

Finally, if we truly hope to attack Conjecture 3.5, we would want to break a divisor $D \in |L|$ into the sum $D \sim D^{\mathrm{vert}} + D^{\mathrm{base}}$, where D^{vert} is a uniruled divisor which is vertical with respect to the Lagrangian fibration, and D^{base} is a uniruled divisor which is pulled back from the base.

Question 3.7. *Can we find a special uniruled decomposition $D \sim D^{\mathrm{vert}} + D^{\mathrm{base}}$ which deforms to a uniruled divisor under general deformations?*

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